School of Electronic Engineering and Computer Science

EBU6408 Sensors and Radio Frequency Identification (RFID)

Lecture 2: RFID Components
Part 2: Tags

Dr Fatma Benkhelifa

PhD, Lecturer (Assistant Professor), SIEEE, Editor of IEEE Communications Letters

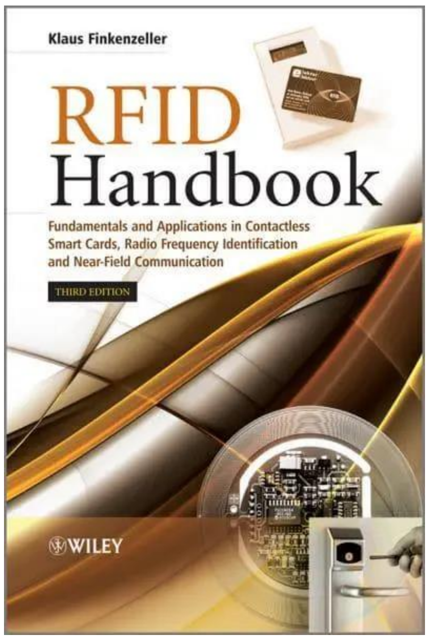
e-mail: *f.benkhelifa@qmul.ac.uk*

website: *Fatma Benkhelifa - EECS (qmul.ac.uk)*

Queen Mary
University of London

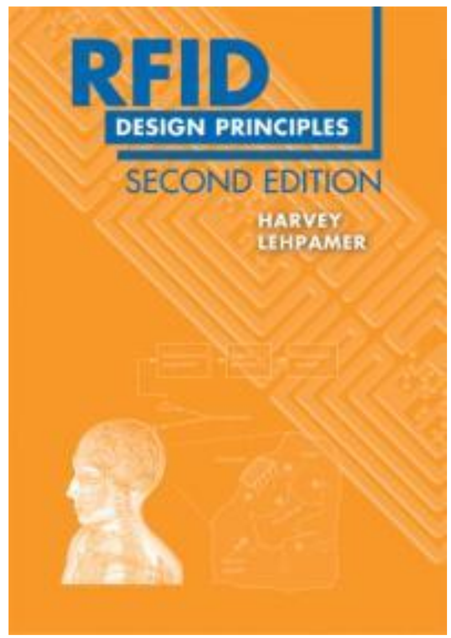
Textbooks

Klaus Finkenzeller, "RFID Handbook", Wiley, Third Edition, 2010.



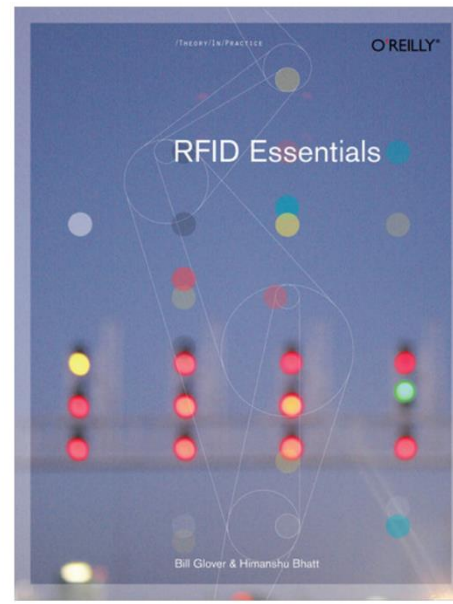
[PDF File]

Harvey Lehpamer, "RFID Design Principles", Artech House, 2012.



QMUL Library link

Bill Glover and Himanshu Bhatt, "RFID Essentials", O'Reilly Media Inc, 2006.



QMUL Library link

Evaluation Rules???

- **Two class tests:**
  - **Test 1:** Covering the material delivered in the first half o module **(5%)**
  - **Test 2:** covering the material delivered in the second half of module **(10%)**
  - **Allowed: : Handwriting** double-faced A4-sized cheating sheet
- **Coursework:** Work in groups of 4/5 students.
  - **Lab 1:** Introduction to Arduino **(Bonus Mark)**
    - ✓ Submission of written report on hardware design and programming code.
  - **Lab 2:** RFID projects using Arduino kits **(5%)**
    - ✓ Submission of written report on hardware design and programming code and **5-min video demonstration**
  - **Mini-project:** using practical knowledge from labs and chatbot support **(10%)**
    - ✓ Submission of written report on hardware design and programming code and **10-min video demonstration**
- Final exam: covering all teaching weeks **(70%)**
  - **Allowed: Handwriting** double-faced A4-sized cheating sheet
- A couple of QM+ online quizzes on teaching lectures **(Bonus Mark)**

Module Activities

Week 1
- Starting 30 March
- Lectures 1-2
- Tutorial 1: last 45 mins

Week 2
- Starting 6 April
- Lectures 3-4
- Tutorial 2: last 45 mins
- Class Test 1 (Date TBD)

Week 3
- Starting 11 May
- Lectures 5-6
- Tutorial 3: last 45 mins

Week 4
- Starting 18 May
- Lectures 7-8
- Tutorial 4: : last 45 mins
- Mock Exam
- Class Test 2 (Date TBD)

- Labs (Arduino):
  - ➢ Two lab sessions will take place in between the teaching weeks.
  - ➢ Work in groups (4/5 students) to share the hardware
  - ➢ Assignment submission deadlines: usually after lab session.
  - ➢ Mini-project due at the end of the term.
- Exact dates will be confirmed later on.

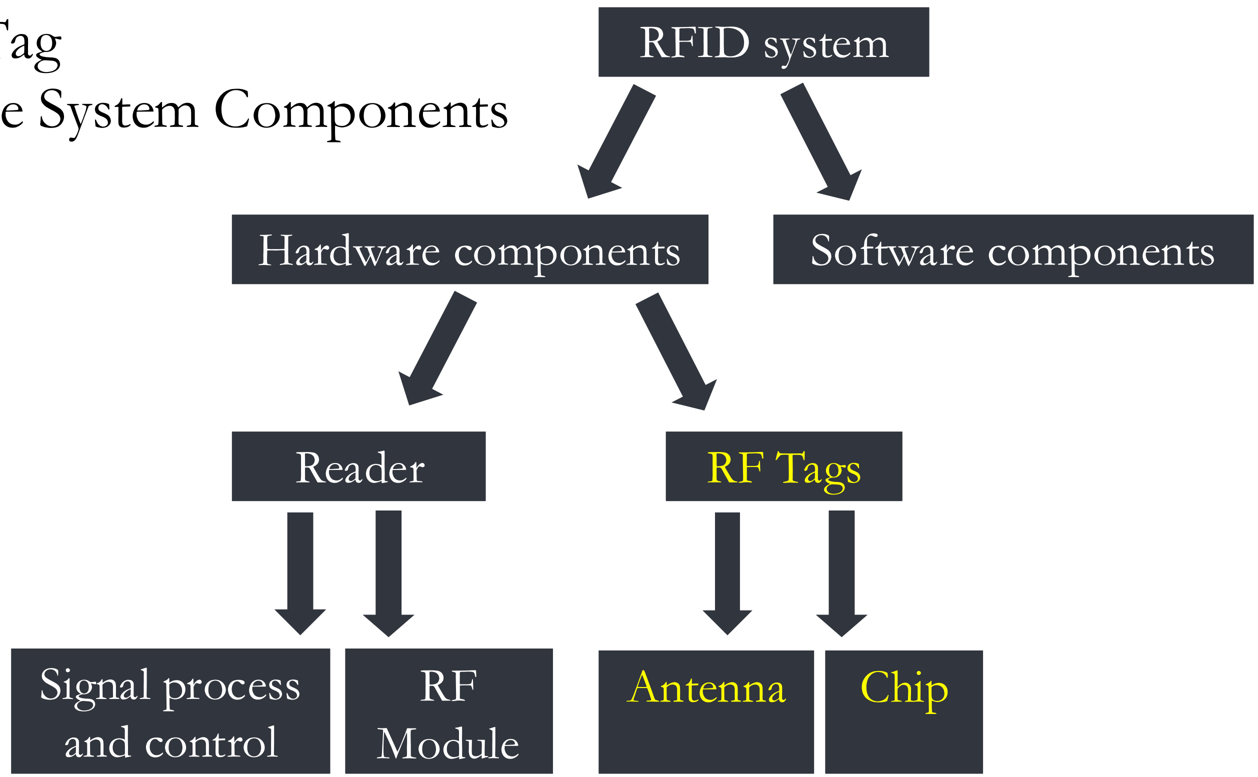
QM+ Quizzes will take place during term time for bonus marks

Module Content

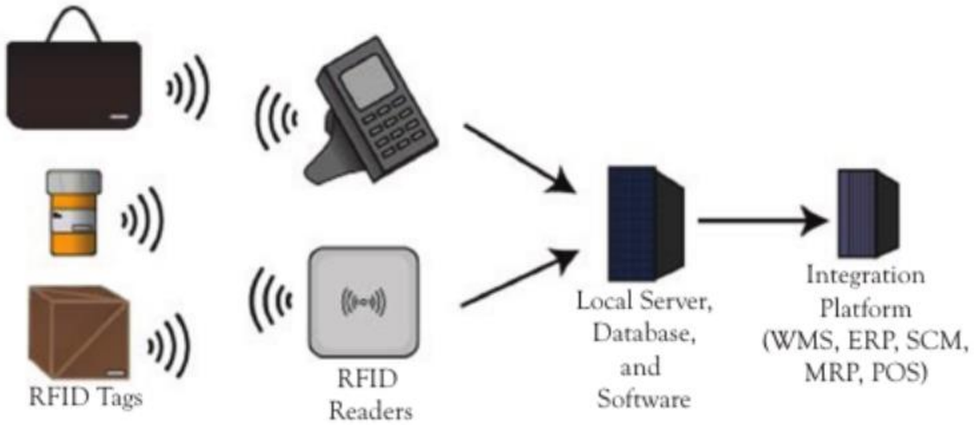
Parts	Lectures	Teaching Weeks	
I. RFID Basics	Lecture 1: RFID Introduction and Overview	Week 1	Test 1
	Lecture 2: RFID System Components		
II. Transmission Modes	Lecture 3: Wireless Communication principle	Week 2	
	Lecture 4: Checksum, Coding and Modulation		
III. Protocol	Lecture 5: RFID Anti-Collision Protocols	Week 3	Test 2
	Lecture 6: EPCGlobal Standard		
IV. System Design	Lecture 7: Key factors of system design	Week 4	
V. Application & Practice	Lecture 8: Application pattern of RFID		
	Lecture 9: Testing and analysis (self-study)		

Outline

- Reader
- RFID Tag
- Software System Components



Operating Principle of RFID Systems



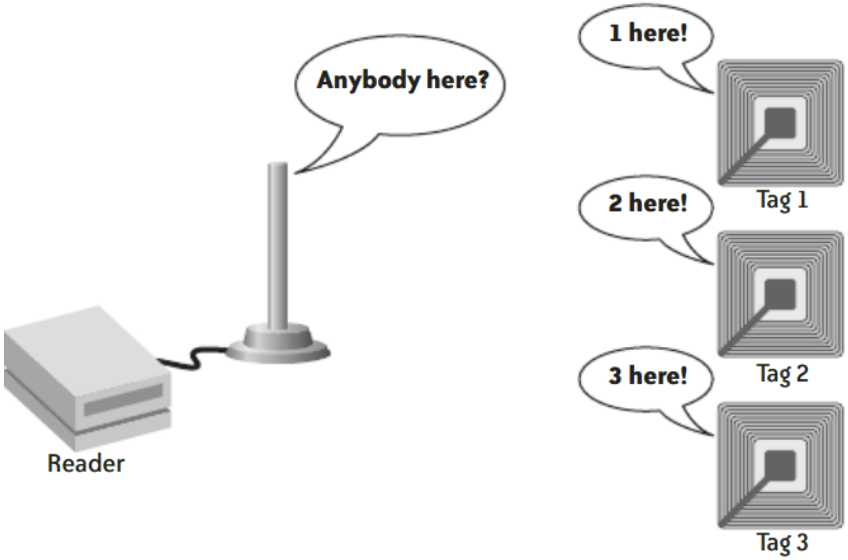
- RFID typically consists of
  - A computer
  - A database
  - RFID software
  - A reader, which receives signals from the RFID and identifies and processes the information contained in one of more tags
  - RFID middleware
  - an antenna, used to transmit signals between the reader and the RFID tag.
  - An RFID tag (transponders), which contains the data of the element to identify
- The different for analysed designations of the tags and readers may vary, depending on applications, contexts, and their hardware and software resources.
- The readers are themselves connected to servers for data processing at middleware and application levels is, archiving and traceability

RFID TAGS

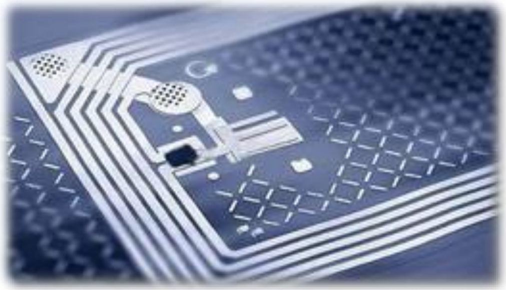
Queen Mary
University of London

Tags

- RFID reader transmits radio signals at a preset frequency and interval (usually hundreds of times every second).
- Any radio frequency tags that are in the reader range will pick up its transmission because each has a **built-in antenna** that is capable of listening to radio signals at a preset frequency.
- The tags use energy from the reader's signal to reflect this signal back.
- Tags may modulate the signal to send information, such as an ID number, back to the reader



RFID Tags



RF tags are significant hardware component in RFID system, which will be introduced from 8 aspects:

Tag Function	Tag Antenna
Tag Classification	Tag Chip
Tag Operational Specifications	Tag Wake-up Circuit
Tag Components	Tag Manufacture

Tag Function

The main function of RF tags is to store a certain amount of data, and send it to the reader in a contactless manner. Generally speaking, tags' functions are concluded as:

Data Storage

Tags store goods-related information, e.g. identifier, production date, manufacturer, etc.

Energy Harvesting

Tags can absorb energy from electromagnetic field generated by the reader, and generate electricity for themselves.

Contactless R&W

Tags can be identified from a certain distance to the reader.

Security Encryption

Collision Concessions

…

Selecting Tags

- Many considerations are involved in selecting tags. They include the following:

1. **Required read range:** Active tags provide a longer read range than passive tags. For retail applications, the read ranges offered by passive tags are usually sufficient.
2. **Material and packaging:** Different materials have differing RF characteristics. For example, liquids may impede the flow of radio waves. Metal containers also pose interference challenges to the readers.
3. **Form factors:** RFID tags come in different sizes. The form factor for the tags used for individual products will depend on the packaging used for these products.
4. **Standards compliance:** It is important to consider whether most of the readers that are generally available will understand the RF tag you select. EPCglobal and the International Standards Organization (ISO) provide standards for communications between RFID tags and readers.
5. **Cost:** The cost of a single RFID tag plays an important role in its selection because most applications use many tags.

Classification by Package Form

Card-like Tag

- Potable, good antenna protection, waterproof

Label-like Tag

- Directly attached to the items, used in industrial production and logistics management

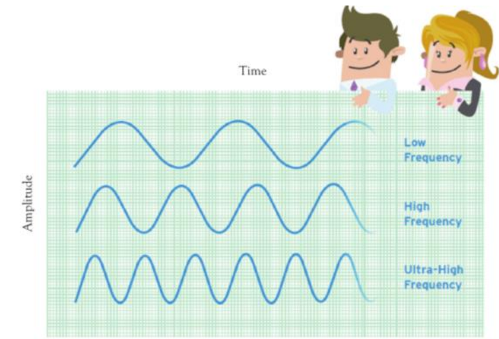
Implantable Tag

- Animal and plant management

Accessories-like Tag

- Easy to carry, no influence on the appearance

Classification by Work Freque



LF Tags

- 30kHz-300kHz
- Usually passive
- Inductively coupled
- Strong penetration

HF Tags

- 300kHz-30MHz
- Usually passive
- Inductively coupled
- Fast transmission rate, large storage

UHF Tags

- 0.3GHz-3GHz
- Passive or active
- Electromagnetic backscatter coupled
- Long distance, high speed, mobile scenario, good multiple-tag r&w performance

Classification by Read/Write Capability

Read-only Tags

- Only readable, unique serial number
- Low price
- Simple structure

Read & Write Tags

- Readable, writable
- Complex structure
- High cost
- No database related

Classification by Power Source

Active Tags

- Rely on its own battery

Passive Tags

- Rely on the carrier signal sent from reflection reader to gain energy

Semi-passive Tags

- The battery is integrated on the board, but as an auxiliary backup

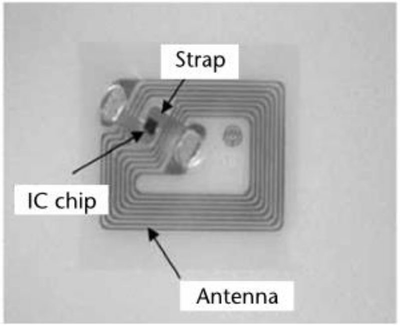
Passive Tags

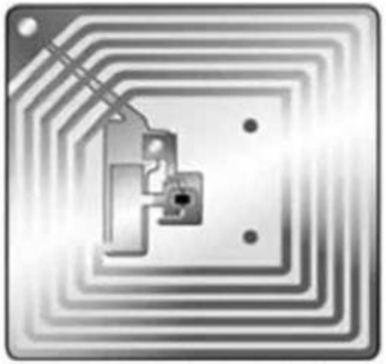
- **No power supply:** electrical current transmitted by the RFID reader inductively powers the device, which allows it to transmit its information back.
- **Limited transmission capability:** typically, no more than an ID number.
- **Limited range of broadcast:** requiring the reader be significantly closer than an active one would.
- **Ideal for places preventing battery replacement**.

❖ Applications: include inventory, product shipping and tracking, hospitals and other medical purposes, and antitheft, can be used implanted into the human body or placed under the skin.

Passive Tags (Cont'D)

- Tags consist of a silicon device (chip) and antenna circuit.
  - Purpose of antenna circuit: induces an energizing signal and sends a modulated RF signal.
  - Read range: largely depends upon the antenna circuit and size.
- The antenna circuit is made of a LC resonant circuit or E-field dipole antenna, depending on the carrier frequency.
  - The LC resonant circuit is used **for frequencies of less than 100 MHz**. At these frequencies, the communication between the reader and tag is achieved with magnetic coupling between the two antennas. An antenna utilizing inductive coupling is often called a **magnetic dipole antenna**.
- The antenna circuits must be designed in such a way to **maximize the magnetic coupling** between them:
  - The LC circuit must be tuned to the carrier frequency of the reader.
  - The Q of the tuned circuit must be maximized.
  - The antenna size must be maximized within physical limit of application requirement.



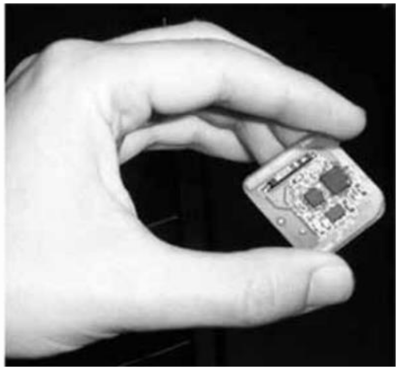


Passive Tags (Cont'D)

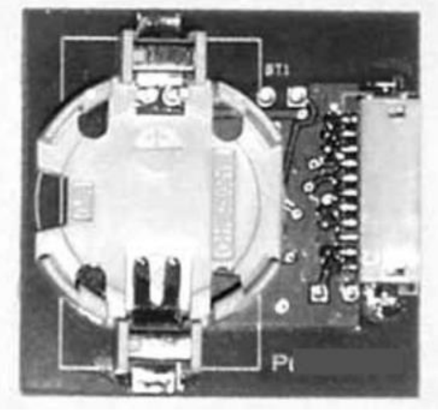
- **Backscattering** of the carrier frequency for sending data from tag to reader.
  - The amplitude of backscattering signal is modulated with the tag's data.
  - Modulation data: ASK (NRZ or Manchester), FSK, or PSK.
- During backscatter modulation, the incoming RF carrier signal to the tag is loaded and unloaded, causing amplitude modulation of the carrier, corresponding to the tag data bits.
- The RF voltage induced in the tag's antenna is amplitude modulated by the modulation signal (data) of the tag device.
- Amplitude modulation can be achieved by using a **modulation transistor** across the LC resonant circuit or partially across resonant circuit.
- Changes in the voltage amplitude of the tag's antenna can affect the voltage of the reader antenna.
  - By monitoring the changes in the reader antenna voltage (due to the tag's modulation data), the data in the tag can be reconstructed.
  - The RF voltage link between the reader and tag antennas are often compared to weakly coupled transformer coils; as the secondary winding, tag coil, is momentarily shunted, the primary winding, reader coil, experiences a momentary voltage change.

Active Tags

- Active tags, also called **transponders:**
  - **Performs a specialized task:** contain a transmitter that is always on and designed for communications up to 30m from the RFID reader.
  - **Powered by on-board power source:** usually a battery about the size of a coin. It does not require inductions to provide current, as is true of the passive tags.
  - **Designed with a variety of specialized electronics:** including microprocessors, different types of sensors, or I/O devices. Depending on the target function of the tag, this information can be processed and stored for immediate or later retrieval by a reader.
  - **Larger and more expensive:** than passive RFID tags. The additional space taken up by a battery
  - **Hold more data about the product:** commonly used for high-value asset tracking. A feature that most active tags have, and most passive tags do not is the ability to store data received from a transceiver.
  - **Ideal in environments with electromagnetic interference:** since they can broadcast a stronger signal in situations that require a greater distance between the tag and the transmitter.



Front Side



Reverse Side

Active and Passive Tags Comparison

- Active and passive RFID technologies are fundamentally distinct technologies with substantially different capabilities.
- **For supply chain asset management applications:** the most effective and complete solution **combines** the advantages of each technology in complementary ways.

1. **Passive RFID:** is most appropriate where the movement of tagged assets is **highly consistent and controlled**, and **little or no security or sensing capability or data storage** is required.

✓ They contain significantly more sophistication, data management, and security concerns.

✓ Cost typically well below $1.

✓ Small as 0.4 mm square and thinner than a sheet of paper.

2. **Active RFID:** is best suited where business processes are **dynamic or unconstrained**, **movement of tagged assets** is variable, and **more sophisticated security, sensing, and/or data storage capabilities** are required.

✓ Generally, cost from $10 to $50, depending on the amount of memory, the battery life required, any on-board sensors, and the ruggedness.

✓ Small as a coin, which means that they are around 50 times the size of passive ones.

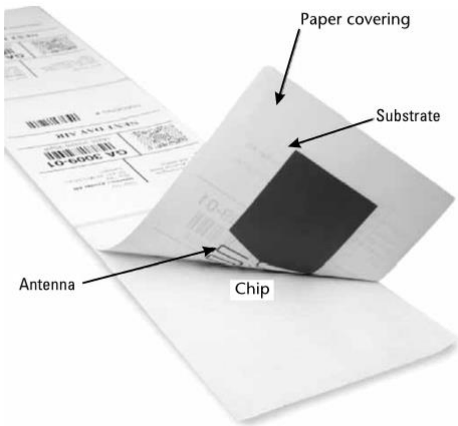
Tag Comparison

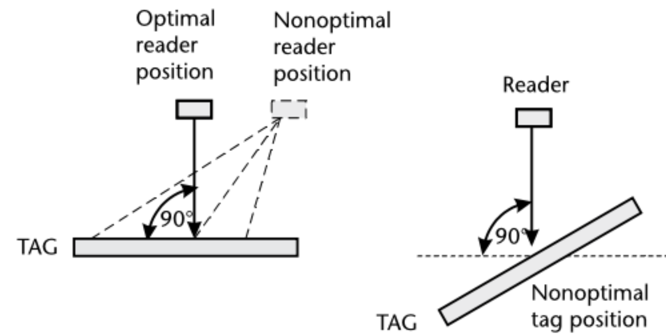
Tag Type	*Power Source*	*Memory*	*Communication Range*
Active	Battery	Most	Greatest
Semi-Passive	Battery and Reader	Moderate	Moderate
Passive	Reader	Least	Least

Tag Operational Specification

Similar to the reader, in practice, several factors are considered in tags' operational specification. Here are a few to be noticed:

Frequency Band	**Size and Form**
Read Range	**EIRP**
Surrounding	**Antenna Polarization Direction**
Applications with Mobility	**Cost**
Reliability	**Power Source**





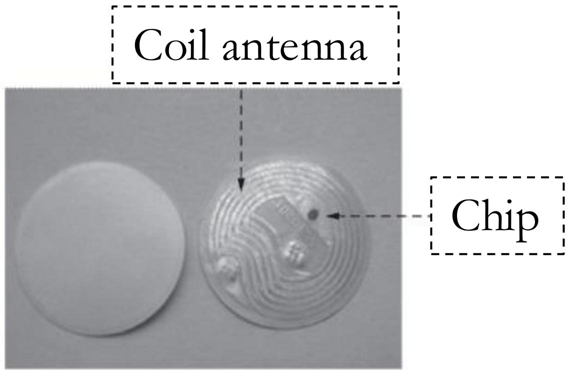
Tag Components

Antenna

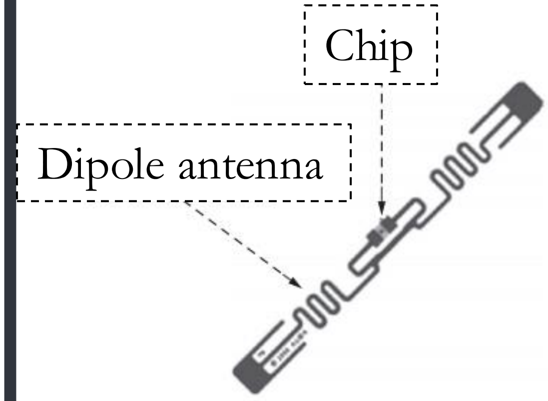
Decides the size of tag.
Receive RF signal from the reader;
Send chip data to the reader;
Provide energy for passive tags

Chip

Demodulate and decode signal received by the antenna, encode and modulate the signal to send from the tag, and perform anti-collision algorithm and store data.

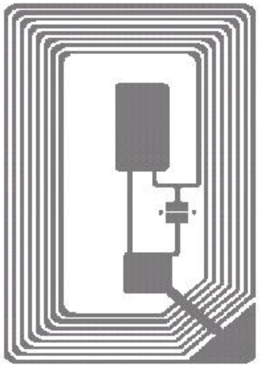


(a)高频标签(13.56MHz)



(b)超高频标签(915MHz)

Tag Antenna Types (Cont'D)



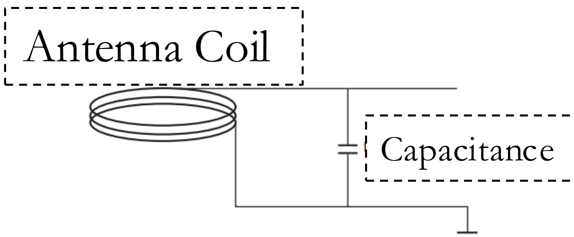
An antenna is a conductive structure specifically designed to couple or radiate electromagnetic energy.

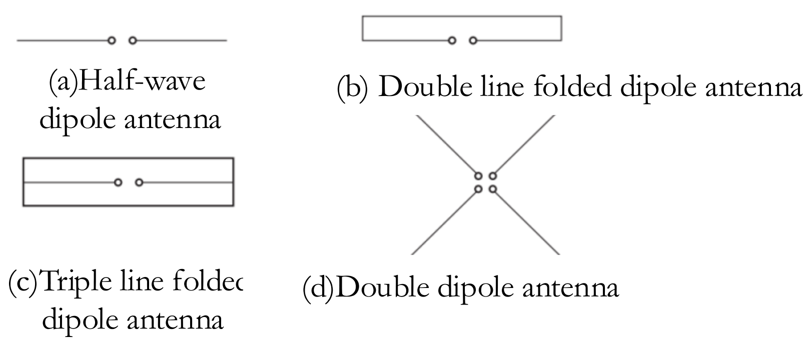
Generally, the smaller the tag size, the less the antenna radiation impedance, the shorter the tag work range, the lower the work efficiency.

Antenna performance includes direction property, antenna efficiency, antenna gain, etc.

According to work principle, tag antenna has 3 types:

Coil antenna	Microstrip patch antenna	Dipole antenna



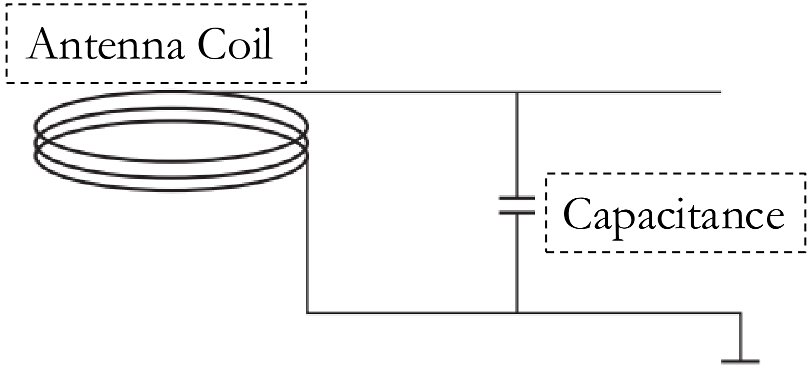


Coil Antenna

Simple process, low cost, widely used in LF, HF close-range RFID system.

Work by inductively coupled, whose principle similar to transformer. The reader antenna is equivalent to transformer primary coil, tag antenna as secondary coil, generating voltage at a varying magnetic field.

Charged by parallel capacitors, generate power for tag chip.

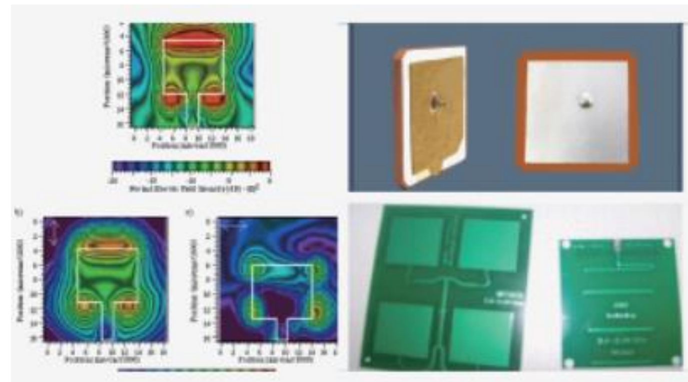


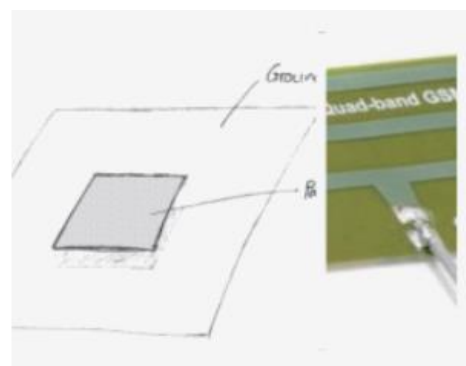
Microstrip Patch Antenna

A microstrip patch antenna is usually attached to the surface of a conductor sheet, with a metal foil ground plane on the other side.

Light weight, small size, low cost, easy to mass production.

Suitable for scenario of which the communication direction doesn't change.





Dipole Antenna

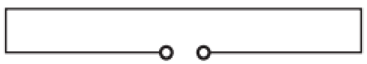
A dipole antenna consists of two straight wires with the same length and thickness put in a straight line.

Suitable for UHF tags.

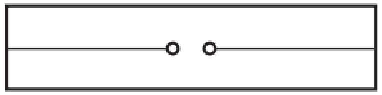
Typical antennas: half-wave / double line folded / triple line folded / double dipole antenna.

A dipole antenna is **omnidirectional** antenna.

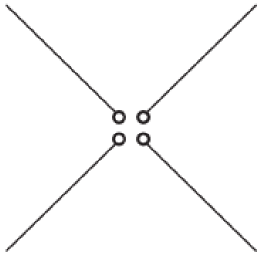
(a)Half-wave dipole antenna



(b)Double line folded dipole antenna



(c)Triple line folded dipole antenna

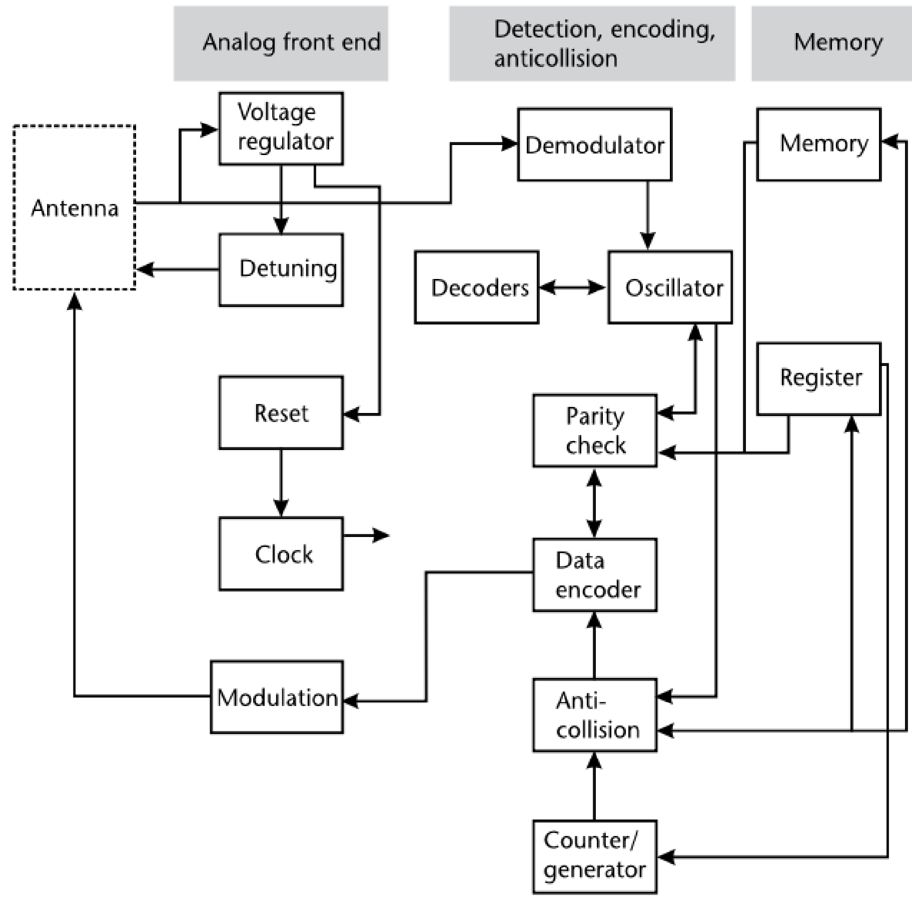


(d)Double dipole antenna

Tag Chip

- **Components:** simulation front end, control unit and storage unit.

1. **Simulation front end:** the rectified antenna input signal provides a stable voltage, the antenna input is detected to obtain a digital signal, modulate control unit signal for antenna to send, provide clock for control unit.
2. **Control unit:** data decode, check, encode, encryption, decryption, anti-collision, read/write control.
3. **Storage unit:** tag data carrier

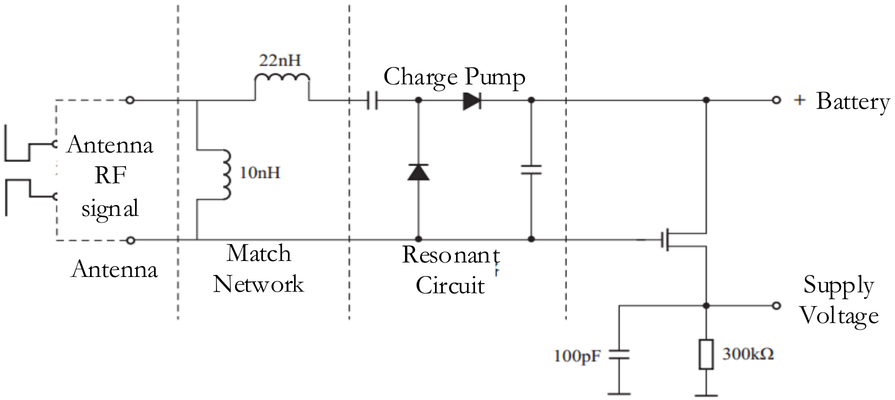


Tag Wake Circuit

Wake mechanism circuit design

Rectifier→1V voltage output →Transistor connection

Rectifier→5mV voltage output →Compare with reference voltage →Transistor connection



Turn-on circuit for the active UHF tag

Tag Manufacturing Process

Main methods of tag manufacture :
Coil winding, chemical etching, printing

Manufacture process:
Antenna produce and chip assembly

Antenna Production Process

Copper foil/ Aluminum foil + PET Substrate → Lamination → Printing Resist ink → Etching → Antenna

Etching antenna manufacture process

Antenna substrate → Printing Conductive ink → Curing ink → Antenna

Printing antenna manufacture process

Antenna substrate → Printing Conductive ink → Curing ink → Plating Thickening → Antenna

Planting antenna manufacture process

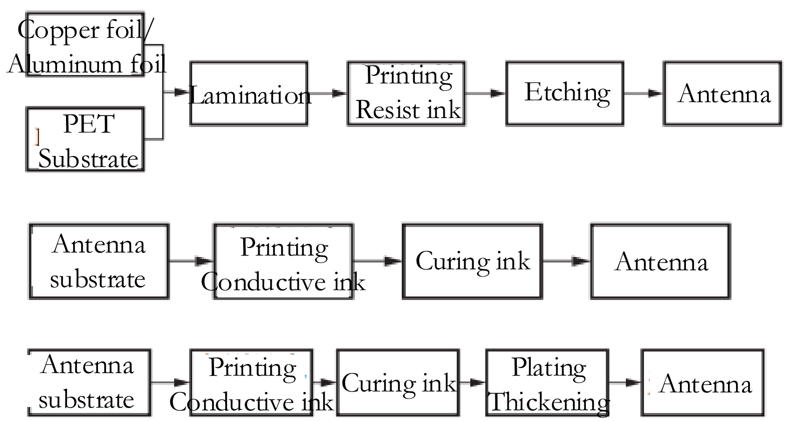
Chip Assembly

Flip chip technology
Fast solid state conductive adhesive dispensing technology

Antenna Production Process

Two methods of antenna production process:

- **Chemical etching:** is an established technology used for the realization of printed electronic circuit boards.
- **Conductive ink printing:** is the process based on **flexographic** equipment, in which the antennas can be printed on a **polymer roll** in a single step with **no need for masking**.



Etching antenna manufacture process

Printing antenna manufacture process

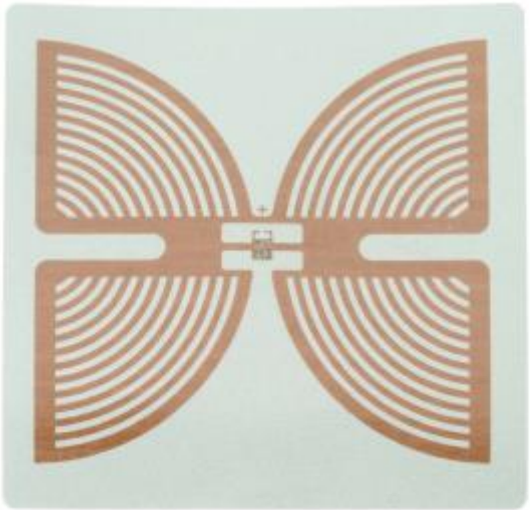
Planting antenna manufacture process

Queen Mary University of London

Chip Assembly

- **Major challenge in assembling:** small size of the chip itself, the **need for low temperature attachment and the required throughput capacity.**
  1. **Standard flip-chip technology:** attaches the chip to the antenna roll **using standard flip-chip technology** and dispensing **fast curing conductive adhesive** at high speed using pick-and-place (PnP) equipment.
     - ✓ quite simple, capable of attaching about 10,000 components per hour, equivalent to about 70 million tags per year.
  2. **Strap attach:** attaches the chip to a **polymeric carrier film** in roll form at high density and high velocity using roll-to-roll equipment.
     - This polymeric carrier has previously been prepared with small caves on the surface to receive the chip, which corresponds to large conductive pads.
     - Once this step is completed, it is possible to couple the roll containing the straps with the roll containing the antennas and accomplish the final assembly by using **dispensed adhesive**.
     - Due to the different density and location of the chip on the strap carrier, each strap needs to be singulated before attachment.
- Advantage of strap attach: possibility to assemble chips at a very high velocity.
- Disadvantage: is that the process consists of **two-step phases** and that the singulation of the strap causes a reduction in the speed of the process.

Example of Tags



Example of passive tag – Intermec Butterfly RFID tag

Example of active tags: RF code Mantis tag

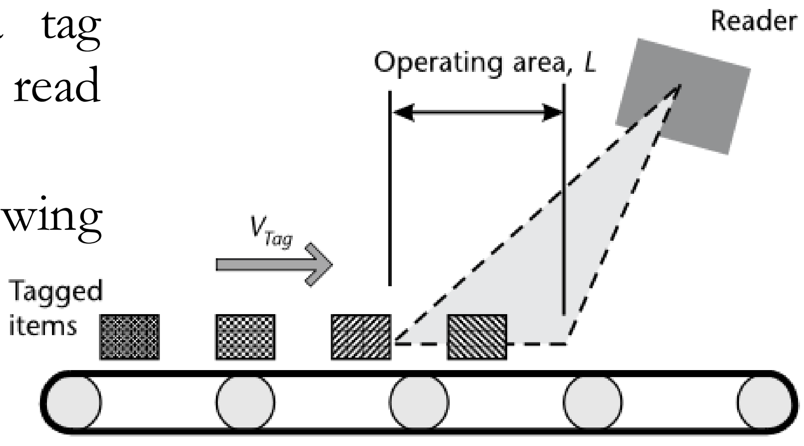
Multiple Tag Operation (Cont'D)

- When containers or freight are moved on a conveyor or similar equipment in a tag reader system, the reader/writer must read and write data to and from moving tags.
- For successful access, the following conditions must be satisfied:

$$T_c = A_{cn} \frac{D_t}{D_r}$$

A_{cn} [count] = average number of tag-reader/writer operations

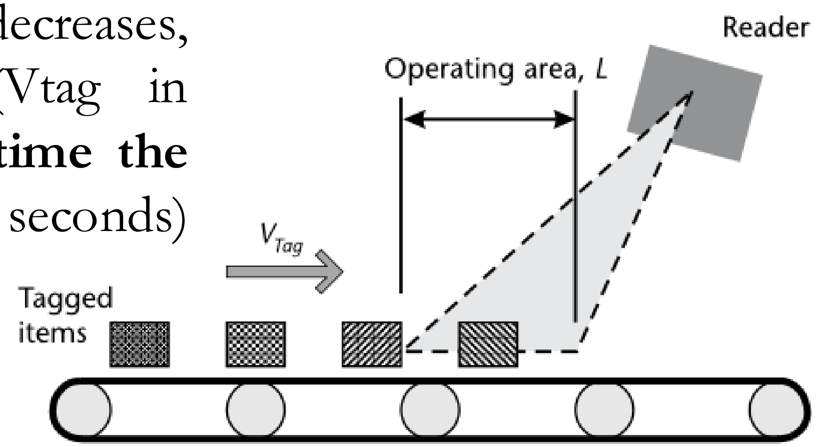
- This formula shows that when **the data transfer volume of the tag** (D_t in bps) increases, and **the data transfer rate** (D_r in bps) decreases, the **tag-reader/writer operation time** (T_c in seconds) increases, and operation may fail.



Multiple Tag Operation (Cont'D)

- When the reader/writer operating area decreases, the distance the tag moves (L in m) decreases, and the tag movement velocity (Vtag in m/second) increases, the **amount of time the tag is in the operating area** (T_r in seconds) decreases, and the operation may fail.

$$T_r = \frac{L}{V_{tag}}$$



- The total amount of time spent in the operating area must be more than the total time taken by the reader/writer and the detection of all tags.

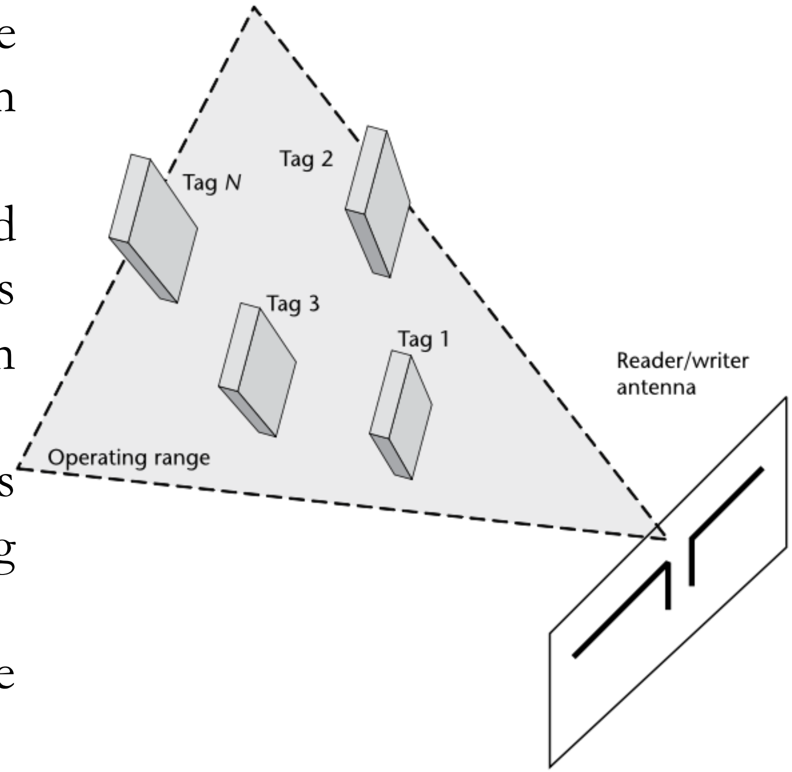
$$T_r \geq T_c + T_d$$

- If only one type of tag can be used when reading/writing RFID tags attached to freight on a conveyor belt, the reader/writer antenna must have a large operating area to cope with the conveyor belt's speed.

Multiple Tag Operation (Cont'D)

- In a multiple-tag operation, the reader/writer must detect the presence of these multiple tags and read/write each of them consecutively.
- This operating method is generally referred to as the ***anticollision protocol*** and is different from the single-tag operation protocol.
- The operating time of multitag operation is several times longer than for single-tag operation.
- The time increases in proportion to the number of tags
- For N_{tag} in the operating area,, the amount of time for which the tags are in the operating area (T_r) is

$$T_r \geq \left(T_c + T_{tag}\right)N_{tag}$$



Multiple Tag Operation (Cont'D)

- In most cases, the tags are **moving**. The reader/writer will generally have to read/write RFID tags **attached to containers or freight being transported on a conveyor belt or trolleys.**
- When T_c is the operating time for a single tag, and T_d is the time required to check for the existence of *N* multitags in an anticollision protocol, the maximum time required for the reader to read/ write all N tags (T_N) is approximated by:

$$T_N = (T_c + T_d)N_{tag}$$

- ***Exp:*** Tag information volume= 16 bytes; Data transfer rate (*Dr*)= 7.8 kbps; $T_c = 0.057\ second$; $T_d = 0.055\ second$; $N = 10$;

$$T_N = (0.057 + 0.055) \times 10 = 1.12\ seconds$$

- Roughly 1.1 seconds are needed for the reader/writer to finish reading/writing all 10 tags.
- If tag information volume is 100 bytes, T_N becomes roughly 7 seconds.
- For the reader/writer to read/write all the tags,

$$T_r \geq T_N$$

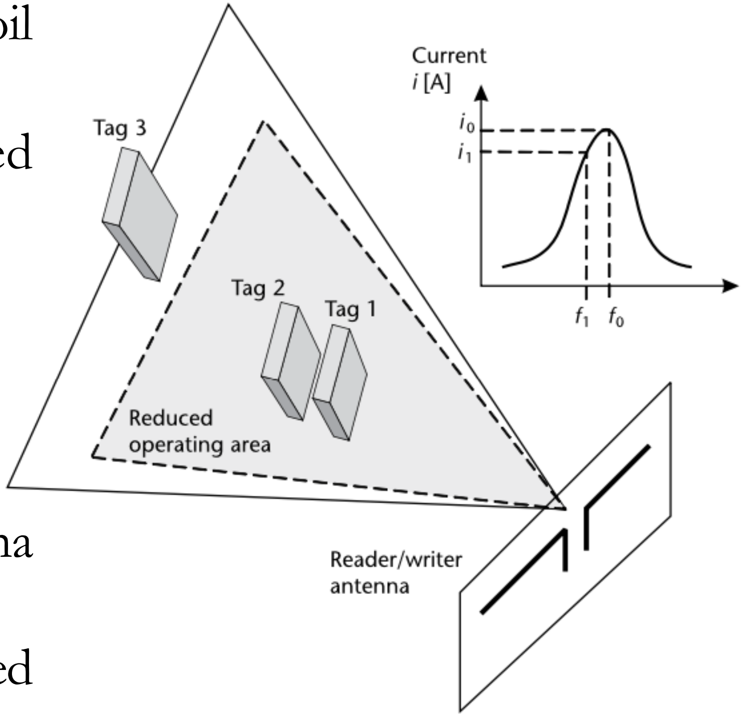
Overlapping Tags

- In inductive frequency band RFID, the resonance characteristic of the tag antenna coil is used for reader/writer operation.
- The tag's resonant frequency, f_0, is calculated by:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

L [H] = inductance of tag antenna coil;
C [F] = capacitance of tag's tuning capacitor.

- If tags overlap, the inductance of their antenna coils is obstructed, and L increases.
- In this case, the resonant frequency expressed by formula becomes lower (f1 < f0).
- As a result, the electromagnetic waves (current i) generated by the tag's coil become **smaller**, and the **operating area decreases**.



RFID POWER SOURCES

RFID Power Sources

- RFID tags need power to sense, compute, and communicate, which is further classified into three categories:
  - Storage: batteries, capacitors.
  - Energy harvesting mechanisms: vibrations/movement, photovoltaic, thermal gradient, and so on.
  - Energy transfer: inductive coupling, capacitive coupling, backscatter.
- Because many of these devices are expected to operate with minimum of human intervention, optimizing power consumption is a very important research area.
- RFID tags may derive the energy to operate either from an on-tag battery or by scavenging power from the electromagnetic radiation emitted by tag readers.
- Storage refers to the way devices store power for their operation done either by using batteries or by using capacitors.
- Batteries are used when a longer life is required and capacitors are used in applications that require energy bursts for very short durations

Power Harvesting Systems

- Power harvesting (sometimes termed energy scavenging) is the process of acquiring energy from the surrounding environment (ambient energy) and converting it into usable electrical energy;
  - **Exp:** the self-winding watch cleverly extracting mechanical energy from the wearer's arm movements.
- The energy transfer is the way by which passive RF devices are powered.
  1. **Inductive coupling:** is the transfer of energy between two electronic circuits due to the mutual inductance between the two circuits.
  2. **Capacitive coupling:** is the transfer of energy between two circuits due to the mutual capacitance between the two circuits.
  3. **Passive backscattering:** is a way of reflecting back the energy from one circuit to another.
- Passive tag is powered through inductive coupling or far-field energy harvesting.

Active Power Sources

- Battery-assisted backscatter tags have their own power source to preenergize the silicon chip. Lithium batteries offer a set of performance and safety characteristics optimal for RFID tag applications. Two types of batteries that are most common are:
  1. **Primary lithium/manganese dioxide ($Li\text{-}MnO_2$):** relatively safe compared to $LiSOCL_2$, does not develop any gas or pressure during battery operation, cannot operate at voltages greater than 3V.
     - At least two Li-MnO2 cells must be connected in series to ensure a proper margin of safety for system reliability.
     - This requirement adds weight and cost while potentially decreasing reliability due to increased part count.
     - Suited for applications that have relatively high continuous or pulse current requirements.
  2. **Lithium-thionyl chloride ($Li\text{-}SOCL_2$):** It has the highest open-circuit voltage of 3.6V.
     - For most applications, only one cell of Li-SOCL 2 is required to maintain sufficient operating voltage.
     - Suited for applications that have relatively high continuous or pulse current requirements.

Other Power Sources

- Power generation from the user may alleviate such design restrictions and may enable new products such as batteryless on-body sensors.
- Power may be recovered passively from body heat, arm motion, typing, and walking or actively through user actions such as winding or pedaling.
- Nuclear power: revealed as possible source of power for wireless sensor and RFID networks.
- It produces huge amount of energy: the energy density measured in mWh/mg is 0.3 for a lithiumion battery, 3 for a methanol-based fuel cell, 850 for a tritium-based nuclear battery, and 57.000 for a polonium-210 nuclear battery.
- The current efficiency of a nuclear battery is around 4%, and current research projects (e.g., as part of the new DARPA program called Radio-Isotope Micropower Sources) aim at 20%.

Other Power Sources

- A novel power supply for implantable biosensors is near infrared light (NIR) transmission which recharges a lithium secondary battery wirelessly through the skin.
- The Sun is a good source of NIR light, and its use requires no other external device to deliver energy to the recharging system.
- A photovoltaic cell array and the rectifier using Schottky diodes implanted beneath the skin can receive NIR light through the skin and charge the battery that is directly powering an implanted biosensor(s).
- In 2008, a U.K.-based consortium of companies has successfully designed and clinically tested an in-body microgenerator that converts energy from the heartbeat into power for implanted medical devices.
- The microgenerator could help power implanted medical devices by augmenting the existing battery for devices such as cardiac pacemakers and implanted defibrillators.
- In preclinical testing, the microgenerator successfully produced one-third of the energy required to power a conventional cardiac pacemaker.

Thank You

Dr Fatma Benkhelifa

Queen Mary
University of London